

Quantitative Discovery of Qualitative Information: A General Purpose Document Clustering Methodology

Gary King

Institute for Quantitative Social Science
Harvard University

Talk at Washington University, St. Louis, 1/21/2010

Joint work with Justin Grimmer, Harvard University

A Method for Conceptualization

- Systematic method for computer-assisted conceptualization from text

A Method for Conceptualization

- Systematic method for computer-assisted conceptualization from text
- Conceptualization through **Classification**: “one of the most central and generic of all our conceptual exercises. . . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).

A Method for Conceptualization

- **Systematic method for computer-assisted conceptualization from text**
- Conceptualization through **Classification**: “one of the most central and generic of all our conceptual exercises. . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).
- We focus on **Cluster Analysis**: simultaneously 1) invent categories and 2) assign documents to categories

A Method for Conceptualization

- Systematic method for computer-assisted conceptualization from text
- Conceptualization through **Classification**: “one of the most central and generic of all our conceptual exercises. . . the foundation not only for conceptualization, language, and speech, but also for mathematics, statistics, and data analysis. . . Without classification, there could be no advanced conceptualization, reasoning, language, data analysis or, for that matter, social science research.” (Bailey, 1994).
- We focus on **Cluster Analysis**: simultaneously 1) invent categories and 2) assign documents to categories
- (We focus on texts, our methods apply more broadly)

Why Johnny Can't Classify (Optimally)

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects
- $Bell(2) = 2$ (AB, A B)

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects
- $Bell(2) = 2$ (AB, A B)
- $Bell(3) = 5$ (ABC, AB C, A BC, AC B, A B C)

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects
- $Bell(2) = 2$ (AB, A B)
- $Bell(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $Bell(5) = 52$

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects
- $Bell(2) = 2$ (AB, A B)
- $Bell(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $Bell(5) = 52$
- $Bell(100) \approx$

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects
- $Bell(2) = 2$ (AB, A B)
- $Bell(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $Bell(5) = 52$
- $Bell(100) \approx 10^{28} \times$ Number of elementary particles in the universe

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects
- $Bell(2) = 2$ (AB, A B)
- $Bell(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $Bell(5) = 52$
- $Bell(100) \approx 10^{28} \times$ Number of elementary particles in the universe
- Now imagine choosing the *optimal* classification scheme by hand!

Why Johnny Can't Classify (Optimally)

- Clustering seems easy; its not!
- $Bell(n)$ = number of ways of partitioning n objects
- $Bell(2) = 2$ (AB, A B)
- $Bell(3) = 5$ (ABC, AB C, A BC, AC B, A B C)
- $Bell(5) = 52$
- $Bell(100) \approx 10^{28} \times$ Number of elementary particles in the universe
- Now imagine choosing the *optimal* classification scheme by hand!
- Its no surprise that automated algorithms can help, but which algorithms?

Why HAL Can't Classify Either

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **unclear**

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **unclear**
 - The literature: **little guidance on when methods apply**

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **unclear**
 - The literature: **little guidance on when methods apply**
 - **Deriving such guidance**: difficult or impossible

Why HAL Can't Classify Either

- Large quantitative literature on **cluster analysis**
- The Goal — an optimal application-independent cluster analysis method — is mathematically impossible:
 - **No free lunch theorem**: every possible clustering method performs equally well on average over all possible substantive applications
- Existing methods:
 - **Many choices**: model-based, subspace, spectral, grid-based, graph-based, fuzzy k -modes, affinity propagation, self-organizing maps,...
 - **Well-defined** statistical, data analytic, or machine learning foundations
 - How to add substantive knowledge: With few exceptions, **unclear**
 - The literature: **little guidance on when methods apply**
 - **Deriving such guidance**: difficult or impossible
- **Deep problem in cluster analysis literature: no way to know which method will work ex ante**

If Ex Ante doesn't work, try Ex Post

If Ex Ante doesn't work, try Ex Post

- Methods and substance must be connected (no free lunch theorem)

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: hard to do it by understanding the model

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: hard to do it by understanding the model
- We do it **ex post** (by qualitative choice). For example:

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: hard to do it by understanding the model
- We do it **ex post** (by qualitative choice). For example:
 - Create long list of clusterings; choose the best

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: hard to do it by understanding the model
- We do it **ex post** (by qualitative choice). For example:
 - Create long list of clusterings; choose the best
 - Too hard for mere humans!

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: hard to do it by understanding the model
- We do it **ex post** (by qualitative choice). For example:
 - Create long list of clusterings; choose the best
 - Too hard for mere humans!
 - An **organized** list will make the search possible

If Ex Ante doesn't work, try Ex Post

- **Methods and substance must be connected** (no free lunch theorem)
- The usual approach fails: hard to do it by understanding the model
- We do it **ex post** (by qualitative choice). For example:
 - Create long list of clusterings; choose the best
 - Too hard for mere humans!
 - An **organized** list will make the search possible
 - E.g.,: consider two clusterings that differ only because one document (of many) moves from category 5 to 6

Our Idea: Meaning Through Geography

Set of clusterings

Our Idea: Meaning Through Geography

Set of clusterings $\approx$
A list of unconnected addresses

wide at **SuperPages.com**

	195	Car	C
37566-1282 28 Allen Ln Ipswich 01938.....	978 356-9960		
81 447-4101 38 Sweet St 02131.....	617 323-7639		
90 257-9961 1 Pleasant St 02131.....	617 442-9780		
37566-1282 Lucilla 124 Harvard Cam 02139.....	617 491-5621		
37564-5188 M 95 Howe St 02133.....	617 323-9713		
361-0380 Mehsin 501 Green Cam 02139.....	617 576-1061		
37566-4548 Corte Nicholas 38 Appleton Boston 02114.....	617 695-6996		
37568-8248 Carter Thos J Sr & Claire 1 Plover Rd MA 02138.....	617 338-9219		
37445-5116 Thomas & Kathleen 50 Thompson Ln MA 02136.....	617 696-6163		
37562-9992 Carter A Sr 02133.....	617 696-6919		
37562-5712 A Huber 31 Bethune Wy Roxbury 02119.....	617 442-1219		
37569-2698 A 200 Putnam Av Cambridge 02142.....	617 492-4174		
37567-5190 A M 255 Muschela Av Bos 02115.....	617 266-7153		
37569-1417 Adams 381 Centre St MA 02186.....	617 698-9074		
37538-1110 Alice 138 Elmwood St 02131.....	617 425-0193		
37562-9195 Andre F 42 Neal St Som 02143.....	617 945-2711		
37562-9195 Carter Anne MD 1161 Beacon St 02446.....	617 739-1022		
37562-9195 Carter Adhans 372 Newbury Boston 02134.....	617 536-6329		
37562-9195 B E 48 Graduate Av Mat 02154.....	617 296-6911		
37562-9195 Carter Barbara L MD Tufts New England Medical Center Bos 02111.....	617 636-9051		
37562-9195 Carter Becky St 02134.....	617 523-4368		
37562-9195 Bernard J 112 Cambridge F Bos 02138.....	617 567-3430		
37562-9195 Bithiah 25 Melrose Dr 02134.....	617 299-8713		
37562-9195 Carter Broadcasting Co 26 Park Plt Bos 02114.....	617 423-9231		
37562-9195 Carter & Burgess Consultants Inc 73 East St Cam 02141.....	617 225-0200		
37562-9195 Carter C 200 Cambridge Av Bos 02135.....	617 782-2118		
37562-9195 C 219 Concord Av East Boston 02128.....	617 569-1545		
37562-9195 C 259 Harvard Cam 02138.....	617 491-4822		
37562-9195 C 218 Mt Vernon St 02124.....	617 226-4392		
37562-9195 C & M 41 Burroughs Jan 02138.....	617 524-9559		
37562-9195 Carter F 24 Hillock St 02131.....	617 327-1105		
37562-9195 Faye & Ricky 107 Cambridge Av Bos 02136.....	617 437-7331		
37562-9195 Francis S 134 Temple W Av Bos 02132.....	617 323-6781		
37562-9195 Franklin & Anne 251 Mt Auburn Cam 02138.....	617 354-0798		
37562-9195 Fred 42 Harvard Cam 02136.....	617 524-3078		
37562-9195 Fred 96 Harvard Av MA 02186.....	617 698-1343		
37562-9195 G & R 8 Myer Den 02104.....	617 436-8906		
37562-9195 G T 27 Franklin Av Som 02145.....	617 623-7121		
37562-9195 Gayle 25 Providence Dr 02134.....	617 825-0322		
37562-9195 Geo S 115 Mus Hill Rd Jan 02138.....	617 522-3215		
37562-9195 George 105 Nathan Bos 02114.....	617 367-9548		
37562-9195 Carter Halliday Associate 107 S Street Bos 02111.....	617 456-1689		
37562-9195 Carter Harry F 26 Baum St Rd W Av Bos 02132.....	617 325-5465		
37562-9195 Carter Hide Co Inc 144 Sumner Bos 02113.....	617 542-7987		
37562-9195 Carter Hilary 41 Harvey Cam 02148.....	617 876-2750		
37562-9195 Horace 361 Walnut Av Roxbury 02119.....	617 442-5307		
37562-9195 Howard Jr 28 Neve One Bos 02116.....	617 445-5552		
37562-9195 J Cam 41 Chatham Ave 02446.....	617 232-7990		
37562-9195 J 518 Harvard Bos 02446.....	617 730-9483		
37562-9195 J 775 Wy Young Wy Roxbury 02132.....	617 323-5374		
37562-9195 Carter J Jacques MD 1 Brookline Pl Bos 02446.....	617 735-8787		
37562-9195 Carter J M 3410 Columbia St Bos 02137.....	617 464-1040		
37562-9195 Carter J M Ornamental Ironworks Pembroke Tpk 02147.....	617 436-5353		
37562-9195 Carter J Veal Co 40 Newbury St 02138.....	617 442-1775		
37562-9195 Carter James 1573 Cambridge St Cam 02138.....	617 492-1214		
37562-9195 James 32 Fisher Av Roxbury 02126.....	617 739-2193		
37562-9195 J 31 Gold Star Rd Cambridge 02148.....	617 876-8841		
37562-9195 Jas L 34 Roslindale Rd MA 02126.....	617 361-0773		
37562-9195 Jas L 14 Adams Rd Newton 02458.....	617 564-0435		
37562-9195 Jeffrey 40 Waver Av Bos 02136.....	617 426-5994		
37562-9195 John 11 Marlboro St 02134.....	617 987-2163		
37562-9195 John 107 Sumner Bos 02131.....	617 423-4334		
37562-9195 John 40 Westwood Rd 02125.....	617 282-1235		
37562-9195 June O 129 A Summit Av Bos 02138.....	617 734-6109		
37562-9195 K 280 Beverly Av Cambridge 02142.....	617 265-9456		
37562-9195 K 17 Exford Dorchester 02122.....	617 282-1593		
37562-9195 Carter Nellie E 323 Massachusetts Av Bos 02135.....	617 267-6483		
37562-9195 Nicholas S F 115 Randolph Av MA 02186.....	617 698-5307		
37562-9195 Nick 21 Gayfield St 02114.....	617 267-5222		
37562-9195 Nick & Debbi 156 Vernick Rd Newton 02459.....	617 527-0480		
37562-9195 Nicole 38 Chickarewell Dr 02125.....	617 822-1203		
37562-9195 P 40 Cranston Pl Bos 02135.....	617 427-4754		
37562-9195 P E 501 E South St Bos 02337.....	617 268-8213		
37562-9195 P L 44 Matthews Bos 02131.....	617 427-9170		
37562-9195 P P 81 Boyer Jan 02134.....	617 968-8692		
37562-9195 Paul & Constance 114 Aspen Av W Bos 02130.....	617 325-2036		
37562-9195 Paul E 501 E South St Bos 02337.....	617 268-4546		
37562-9195 Paul M 27 Union St 02135.....	617 787-2115		
37562-9195 Carter Pike Driving Inc 27 Beaver Ct Framingham 01702.....	508 876-7811		
37562-9195 Carter Prudence 40 Franklin Woburn 02127.....	617 393-3782		
37562-9195 Prudence 40 Franklin Woburn 02127.....	617 926-7063		
37562-9195 Reginald 106 Brookwood Dorchester 02215.....	617 541-2843		
37562-9195 Renee & Andrew 10 Walnut Bos 02108.....	617 720-3765		
37562-9195 Carter Rice Dowd Baker Dennis Publishing 163 Main Woburn 01887.....	800 638-1671		
37562-9195 Toll Free-Dial 'I' & Then.....	800 619-7447		
37562-9195 Carter Richard 101 Free-Dial 'I' & Then.....	800 648-7447		
37562-9195 Carter Richard 101 Free-Dial 'I' & Then.....	978 988-7447		
37562-9195 Carter Richard 101 Free-Dial 'I' & Then.....	800 638-1673		
37562-9195 Carter Richard 101 Free-Dial 'I' & Then.....	617 987-0823		
37562-9195 Carter Richard A 207 Cambridge Av Brighton 02109.....	617 864-1535		
37562-9195 Carter Richard A 170 Cambridge Av Bos 02136.....	617 267-0710		
37562-9195 Carter Richard K 13 Merwin St 02137.....	617 268-9448		
37562-9195 C 175 Melrose Av Cam 02142.....	617 864-1535		
37562-9195 Rogers 150 St Pauls Bos 02131.....	617 424-6148		
37562-9195 Roy 41 Concord St 02126.....	617 491-6115		
37562-9195 Royce 18 Sumner Cir 02129.....	617 241-0418		

Our Idea: Meaning Through Geography

Set of clusters $\approx$

A list of unconnected addresses

wide at **SuperPages.com**

	195	Car	C
Cartage New England Inc 26 Allen Ln Ipswich 01938.....	978 356-9960		
Cartagena Lydia 38 Sweet Rd 02131.....	617 323-7639		
Cartagena Aveth 1 Fairview Rd 02139.....	617 442-9780		
8 Hrd 02138.....	617 361-5253		
Justicia 50 Decatur Cha 02129.....	617 241-0152		
Lucilla 124 Harvard Cam 02139.....	617 491-5621		
M 95 New Rd 02133.....	617 323-9713		
Melvin 501 Green Cam 02139.....	617 576-1061		
Carte Nicholas 18 Aspen Road 02134.....	617 695-6996		
Carteseno O 4 Harvard Cha 02131.....	617 338-9219		
Carten Thos J Sr & Claire 1 Ivesdale Rd 02138.....	617 698-6163		
Thomas & Kathleen 50 Thompson Ln 02136.....	617 696-6919		
Carte A 100 02133.....	617 329-2257		
A Huber 31 Bethune Wy 02139.....	617 442-5230		
A 200 Putnam Av Cambridge 02139.....	617 492-4174		
A M 255 Metchua Av 02135.....	617 266-7153		
Adams 361 Centre St 02138.....	617 698-9074		
Allice 108 Elmwood Rd 02134.....	617 425-0193		
Allice 40 Market Cambridge 02139.....	617 945-2711		
Andrew F 42 Wal St 02133.....	617 625-6235		
Carte Anne MD 1161 Beacon W 02144.....	617 739-1022		
Carte 196-1593 272 Newbury Street 02134.....	617 536-6329		
B E 18 48 Graduate Av 02134.....	617 296-6911		
Cartier Barbara L MD Turks New England Medical Center 02131.....	617 636-9051		
Cartier Becky 02134.....	617 523-4368		
Cartier Adhene Bernard J 127 Cambridge St 02138.....	617 567-3430		
Bithiah 25 Midway Av 02137.....	617 298-8713		
Bishop 25 Midway Av 02138.....	617 367-9931		
Cartier Broadcasting Co 26 Park Pl 02134.....	617 423-0210		
Cartier C Burgess Consultants Inc 73 2nd St 02134.....	617 225-0200		
Cartier C 2000 Cambridge Av 02135.....	617 782-2118		
C 210 Harvard Av East Boston 02128.....	617 549-1545		
C 109 Harvard Cam 02138.....	617 491-4822		
C 109 Harvard Cam 02138.....	617 296-4392		
C & M 41 Burroughs Jan 02138.....	617 524-5595		
Carter F 24 Hillock Rd 02131.....	617 327-1105		
Faye & Ricky 107 Columbia Av 02136.....	617 437-7331		
Francis S 134 Temple W Av 02132.....	617 323-6781		
Franklin & Anne 291 Mt Auburn Cam 02138.....	617 354-0798		
Fred 42 Harvard Cam 02136.....	617 524-3078		
Fred 16 Harvard Rd 02138.....	617 698-1343		
G & E 8 Harvard Dev 02134.....	617 436-8906		
G T 27 Fyfield Av 02135.....	617 623-7121		
Gayle 25 Franklin Dev 02134.....	617 825-0322		
Geo S 115 Maple Mt Rd Jan 02138.....	617 522-3215		
George 125 Madison Rd 02131.....	617 367-9548		
Carter Halliday Associate 107 S Street Box 02111.....	617 456-1689		
Carter Harry F 26 Baring Jct Rd W Av 02132.....	617 325-5465		
Carter Hide Co Inc 141 Sumner Ave 02131.....	617 542-7987		
Carter Hilary 41 Harvey Cam 02140.....	617 876-2750		
Horace 361 Walnut Av 02139.....	617 442-5307		
Howard Jr 38 Nebe One Box 02118.....	617 445-5552		
J Cam 41 Chatham Bx 02144.....	617 232-7990		
J 35 Harvard Bx 02144.....	617 730-9483		
J 775 The Young Wind Study 02135.....	617 323-5374		
Carter J Jacques MD 1 Brookline Pl Bx 02144.....	617 735-8787		
Carter J M 3410 Columbia Rd S Box 02137.....	617 464-1040		
Carter M 01 Ornamental Ironworks Pembroke Falls 02136.....	617 436-5353		
Carter J Veal Co 40 Newbury St 02138.....	617 442-1775		
Carter James 1573 Cambridge St Cam 02138.....	617 492-1214		
James 362 Foster Av 02136.....	617 739-2193		
J 31 441 St Rd 02136.....	617 876-8841		
Jas L 34 Broadway Rd 02136.....	617 361-0773		
Janie 14 Adams Rd 02146.....	617 964-0435		
Jeffrey 41 Warren Av 02134.....	617 426-5994		
John 11 Mansfield Rd 02134.....	617 987-2163		
John 207 Sumner Box 02137.....	617 423-4334		
John 40 Westwood Rd 02125.....	617 282-1235		
June O 129 A Sunset Av 02138.....	617 734-6109		
K 17 Broadway Av 02134.....	617 265-9456		
K 17 Concord Road 02123.....	617 282-1593		
Carter Nellie E 323 Mauchess Av Box 02135.....	617 267-6483		
Nicholas S F 115 Randolph Av 02136.....	617 698-5307		
Nick 21 Fairfield Box 02114.....	617 267-5222		
Nick & Debbi 156 Vermont Rd Newton 02459.....	617 527-0480		
Norman G 38 Chickadee Rd 02125.....	617 822-1203		
P 41 Cranston Pl Box 02135.....	617 427-4754		
P E 501 E South St Box 02137.....	617 268-4213		
P L 44 Matthews Box 02131.....	617 427-9170		
P R 1 Boyer Jan 02134.....	617 968-8692		
Paul & Constance 114 Aspen Av W Box 02130.....	617 325-2036		
Paul E 501 E South St Box 02137.....	617 268-4546		
Paul M 27 Union St 02135.....	617 787-2115		
Carter Pile Driving Inc 17 Beaver Ct Franklin 02102.....	617 781-235-0488		
Carter Prudence 40 Franklin Waterbury 02172.....	617 393-3782		
Prudence 40 Franklin Waterbury 02172.....	617 393-3782		
Reginald 106 Brookside Cambridge 02213.....	617 541-2843		
Renee & Andrew 10 Walnut Box 02138.....	617 720-3765		
Carter Rice David Baker Dennis Publishing 163 Main Wilmington 01887 Toll Free 800-1 & 2 Then.....	800 638-1671		
Cost Inc 1600 1st Street 02134 Main Wilmington Toll Free 800-1 & 2 Then.....	800 619-7447		
Toll Free 800-1 & 2 Then.....	800 648-7447		
Headquarters 412 Main Wilmington 01887 Toll Free 800-1 & 2 Then.....	800 988-7447		
Ingalls Crane 363 Main Wilmington 01887 Toll Free 800-1 & 2 Then.....	800 988-1673		
Carter Richard 2075 Carver Av Brighton 02137.....	617 987-0836		
Richard A 974 Vernon Box 02136.....	617 556-7293		
Carter Richard A MD 170 0th Street Box 02136.....	617 987-0710		
Carter Richard K 133 Merwin St 02137.....	617 268-9448		
Robert L 175 Rockdale Av Cam 02140.....	617 864-1535		
Roger 150 St Pauls Box 02131.....	617 424-6148		
Roy 41 Concord Rd 02134.....	617 491-6115		
Royce 185 Salisbury Cha 02129.....	617 241-9418		



Our Idea: Meaning Through Geography

Set of clusterings $\approx$

A list of unconnected addresses

wide at SuperPages.com

195		Car		C	
37 566-1282	Cortage New England Inc	28 Allen Ln Ipswich 01938	978 356-9960		
81 447-4101	Cortagema Lydia	38 Sweet Box 02131	617 323-7639		
90 257-9961	Cortagema Aveth	1 Fairview Rd 02139	617 442-9780		
37 566-1282	Jessica	8 Had 02136	617 361-5253		
37 564-5188	Lucilla	124 Harvard Cam 02129	617 241-0152		
361-0380	M 95 Howe Box 02133	617 491-5621			
	M 95 Howe Box 02133	617 323-9713			
	M 95 Howe Box 02139	617 576-1061			
	Carte Nicholas	38 Appleton Boston 02114	617 695-6996		
37 566-4548	Carlegeno O 448ford Box 02131	617 338-0219			
37 628-8248	Carten Thos J Sr & Claire	1 Ivesdale Rd 02136	617 698-6163		
37 445-5116	Thomas & Kathleen	50 Thompson Ln 02136	617 696-6919		
37 822-9992	Carte A Box 02133	617 339-2257			
37 427-5712	A Huber	617 442-5230			
37 569-2698	A 31 Bethune Wy Rindley 02131	617 442-1219			
37 667-5190	A 200 Putnam Av Cambridge 02140	617 492-4174			
37 569-1417	Adams	301 Centre St 02138	617 698-7074		
37 338-9110	Allice	208 Elmwood Rd 02133	617 425-0193		
	Allice	40 Market Cambridge 02139	617 945-2711		
	Andrew F 42 Wal St 02133	617 625-6253			
37 825-1919	Carte Anne MD	617 739-1022			
37 296-1593	1101 Beacon Hts 02144	617 336-6329			
37 670-2078	B E 18 Gladstone Av 02136	617 296-6911			
37 623-9001	Cartier Barbara L MD	200 New England Medical Center 02111	617 636-9051		
37 296-4725	Cartier Becky Jo 02134	617 523-4368			
37 542-1521	Bernard J	127 Cambridge St 02136	617 567-3430		
37 364-5232	Bithiah	25 Midway Dr 02134	617 298-8713		
37 541-5649	Bishop B 20 Newbury St 02138	617 367-9931			
37 739-2662	Cartier Broadcasting Co	26 Park Pl 02136	617 423-0210		
37 879-0030	Cartier C 2000 Cambridge Av 02135	617 225-0200			
37 936-1511	C 2101 Harvard Av East Boston 02128	617 782-2118			
37 569-4119	C 109 Harvard Cam 02138	617 491-4822			
37 569-4782	C & M 41 Burroughs Jan 02136	617 524-5595			
	Carter F 24 Hilltop Box 02131	617 327-1105			
	Faye & Ricky	107 Columbia Av Box 02136	617 437-7331		
	Francis S 134 Temple W Av 02132	617 323-6781			
	Franklin & Anne	201 Mt Auburn Cam 02138	617 354-0798		
	Fred 42 Harvard Jan 02136	617 524-3078			
	Fred 96 Hawley Rd 02138	617 698-1343			
	G & B 8 Harvard Box 02134	617 436-8906			
	G T 27 Franklin Av Box 02145	617 623-7121			
	Gayle 25 Franklin Box 02134	617 825-0322			
	Geo S 115 Miss Hill Rd Jan 02138	617 522-3215			
	George 125 Madison Box 02131	617 367-9548			
	Carter Halliday Associate	107 S Street Box 02111	617 456-1689		
	Carter Harry F	107 S Street Box 02111	617 325-5465		
	Carter Hide Co Inc	140 Sumner Box 02131	617 542-7987		
	Carter Hilary 41 Harvey Cam 02140	617 876-2750			
	Horace	301 Walnut Av Rindley 02131	617 442-5307		
	Howard Jr 38 Nebe One Box 02118	617 445-5552			
	J Cam	41 Chatham Box 02144	617 334-2658		
	J 31	617 232-7990			
	315 Harvard Box 02144	617 730-9483			
	J 775 The Young Wind Rindley 02136	617 323-5574			
	Carter J Jacques MD	617 735-8787			
	Carter J M	1 Broadview Pl Box 02144	617 735-8787		
	3410 Columbia Rd S Box 02137	617 464-1040			
	Carter J M Ornamental Ironworks	200 Franklin Falls 02136	617 436-5353		
	Carter J Veal Co	40 Newbury St 02138	617 442-1775		
	Carter James	1573 Cambridge St Cam 02136	617 492-1214		
	James	302 Fisher Av Rindley 02136	617 739-2193		
	J 31	617 876-8841			
	Jas L 34 Broadview Rd 02136	617 361-0773			
	James 14 Adams Rd Newton 02458	617 564-0435			
	John 41 Warren Av Box 02134	617 426-5994			
	John 11 Marshall Rd 02134	617 987-2163			
	John 207 Sumner Box 02137	617 423-4334			
	John 40 Westwood Rd Box 02125	617 282-1235			
	June O 129 A Summit Av 02138	617 734-6109			
	K 109 Harvard Cam 02138	617 265-9456			
	K 17 Elwood Cambridge 02137	617 282-1593			
	Carter Nellie E	323 Mauchess Av Box 02135	617 267-6483		
	Nicholas S F	115 Randolph Av 02136	617 698-5307		
	Nick 21 Fairfield Box 02114	617 267-5222			
	Nick & Debbi	156 Vermont Rd Newton 02459	617 527-0480		
	Norman G	38 Chickadee Dr 02125	617 822-1203		
	P 41 Cranford Pl Box 02135	617 427-4754			
	P E 501 E South S Box 02137	617 268-4213			
	P L 44 Matthews Box 02131	617 427-9170			
	P R 81 Boyer Jan 02136	617 983-8692			
	Paul & Constance	114 Adams Av W Box 02130	617 325-2036		
	Paul E 501 E South S Box 02137	617 268-4546			
	Paul M 27 Union St 02135	617 787-2115			
	Carter Pile Driving Inc 17 Beaver Ct	Franklin 02102	617 781-235-0488		
	Carter Prudence	40 Franklin Waterbury 02172	617 393-3782		
	Prudence	40 Franklin Waterbury 02172	617 393-3782		
	Reginald	100 Broadview Cambridge 02215	617 541-2843		
	Renee & Andrew	10 Walnut Box 02138	617 720-3765		
	Carter Rice Dowd	3400 Franklin Falls 02136	617 436-5353		
	Ted Free-Old 'I' & Then	800 638-1671			
	Carl Free-Old 'I' & Then	800 619-7447			
	Ted Free-Old 'I' & Then	800 648-7447			
	Headquarters 412 Main Waterbury 02102	978 988-7447			
	Cal	800 638-1673			
	Ingrida Ores 303 Main Waterbury 01887	978 988-7447			
	Carter Richard	207 Carver Av Brighton 02132	617 987-0836		
	Richard A 9748 Vernon Box 02106	617 566-7293			
	Carter Richard A MD	170 Centre St 02136	617 267-0710		
	Carter Richard M	13 Merwin St Box 02137	617 268-9448		
	Robert L 375 Rockdale Av Cam 02140	617 864-1535			
	Roger 130 St Pauls Box 02131	617 424-6148			
	R 109 Harvard Cam 02138	617 491-6115			
	Royce 38 Salisbury Box 02129	617 241-9418			



$\approx$ We develop a (conceptual) geography of clusterings

A New Strategy

Make it easy to choose best clustering from millions of choices

A New Strategy

Make it easy to choose best clustering from millions of choices

- 1 Code text as numbers (in one *or more* of several ways)

A New Strategy

Make it easy to choose best clustering from millions of choices

- 1 Code text as numbers (in one *or more* of several ways)
- 2 Apply all clustering methods we can find to the data — each representing different (unstated) substantive assumptions (<15 mins)

A New Strategy

Make it easy to choose best clustering from millions of choices

- 1 **Code text as numbers** (in one *or more* of several ways)
- 2 **Apply all clustering methods we can find** to the data — each representing different (unstated) substantive assumptions (<15 mins)
- 3 (Too much for a person to understand, but organization will help)

A New Strategy

Make it easy to choose best clustering from millions of choices

- 1 **Code text as numbers** (in one *or more* of several ways)
- 2 **Apply all clustering methods we can find** to the data — each representing different (unstated) substantive assumptions (<15 mins)
- 3 (Too much for a person to understand, but organization will help)
- 4 Develop an **application-independent distance metric** between clusterings, a **metric space of clusterings**, and a **2-D projection**

A New Strategy

Make it easy to choose best clustering from millions of choices

- 1 **Code text as numbers** (in one *or more* of several ways)
- 2 **Apply all clustering methods we can find** to the data — each representing different (unstated) substantive assumptions (<15 mins)
- 3 (Too much for a person to understand, but organization will help)
- 4 Develop an **application-independent distance metric** between clusterings, a **metric space of clusterings**, and a **2-D projection**
- 5 “**Local cluster ensemble**” creates a new clustering at any point, based on weighted average of nearby clusterings

A New Strategy

Make it easy to choose best clustering from millions of choices

- 1 **Code text as numbers** (in one *or more* of several ways)
- 2 **Apply all clustering methods we can find** to the data — each representing different (unstated) substantive assumptions (<15 mins)
- 3 (Too much for a person to understand, but organization will help)
- 4 Develop an **application-independent distance metric** between clusterings, a **metric space of clusterings**, and a **2-D projection**
- 5 “**Local cluster ensemble**” creates a new clustering at any point, based on weighted average of nearby clusterings
- 6 A new **animated visualization** to explore the space of clusterings (smoothly morphing from one into others)

A New Strategy

Make it easy to choose best clustering from millions of choices

- 1 **Code text as numbers** (in one or more of several ways)
- 2 **Apply all clustering methods we can find** to the data — each representing different (unstated) substantive assumptions (<15 mins)
- 3 (Too much for a person to understand, but organization will help)
- 4 Develop an **application-independent distance metric** between clusterings, a **metric space of clusterings**, and a **2-D projection**
- 5 **“Local cluster ensemble”** creates a new clustering at any point, based on weighted average of nearby clusterings
- 6 A new **animated visualization** to explore the space of clusterings (smoothly morphing from one into others)
- 7 **↪ Millions of clusterings, easily comprehended** (takes about 10-15 minutes to choose a clustering with insight)

You choose one (or more), based on insight, discovery, useful information,...



Application-Independent Distance Metric: Axioms

Application-Independent Distance Metric: Axioms

- Metric based on 3 assumptions

Application-Independent Distance Metric: Axioms

- Metric based on 3 assumptions
 - ① Distance between clusterings: a function of the **pairwise document agreements** (pairwise agreements $\Rightarrow$ triples, quadruples, etc.)

Application-Independent Distance Metric: Axioms

- Metric based on 3 assumptions
 - ① Distance between clusterings: a function of the **pairwise document agreements** (pairwise agreements $\Rightarrow$ triples, quadruples, etc.)
 - ② **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)

Application-Independent Distance Metric: Axioms

- Metric based on 3 assumptions
 - ① Distance between clusterings: a function of the **pairwise document agreements** (pairwise agreements $\Rightarrow$ triples, quadruples, etc.)
 - ② **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)
 - ③ **Scale**: the maximum distance is set to $\log(\text{num clusters})$

Application-Independent Distance Metric: Axioms

- Metric based on 3 assumptions
 - ① Distance between clusterings: a function of the **pairwise document agreements** (pairwise agreements $\Rightarrow$ triples, quadruples, etc.)
 - ② **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)
 - ③ **Scale**: the maximum distance is set to $\log(\text{num clusters})$
- $\rightsquigarrow$ **Only one measure satisfies all three** (the “variation of information”)

Application-Independent Distance Metric: Axioms

- Metric based on 3 assumptions
 - ① Distance between clusterings: a function of the **pairwise document agreements** (pairwise agreements $\Rightarrow$ triples, quadruples, etc.)
 - ② **Invariance**: Distance is invariant to the number of documents (for any fixed number of clusters)
 - ③ **Scale**: the maximum distance is set to $\log(\text{num clusters})$
- $\rightsquigarrow$ **Only one measure satisfies all three** (the “variation of information”)
- Meila (2007): derives same metric using different axioms (lattice theory)

Evaluating the Performance of Our Method

Evaluating the Performance of Our Method

- Goals:

Evaluating the Performance of Our Method

- Goals:
 - **Validate Claim:** computer-assisted conceptualization outperforms human conceptualization

Evaluating the Performance of Our Method

- Goals:
 - **Validate Claim**: computer-assisted conceptualization outperforms human conceptualization
 - **Demonstrate**: new experimental designs for cluster evaluation

Evaluating the Performance of Our Method

- Goals:
 - **Validate Claim**: computer-assisted conceptualization outperforms human conceptualization
 - **Demonstrate**: new experimental designs for cluster evaluation
 - **Inject human judgement**: relying on insights from survey research

Evaluating the Performance of Our Method

- Goals:
 - **Validate Claim**: computer-assisted conceptualization outperforms human conceptualization
 - **Demonstrate**: new experimental designs for cluster evaluation
 - **Inject human judgement**: relying on insights from survey research
- We now present three evaluations

Evaluating the Performance of Our Method

- Goals:
 - **Validate Claim**: computer-assisted conceptualization outperforms human conceptualization
 - **Demonstrate**: new experimental designs for cluster evaluation
 - **Inject human judgement**: relying on insights from survey research
- We now present three evaluations
 - Quality $\Rightarrow$ RA coders

Evaluating the Performance of Our Method

- Goals:
 - **Validate Claim**: computer-assisted conceptualization outperforms human conceptualization
 - **Demonstrate**: new experimental designs for cluster evaluation
 - **Inject human judgement**: relying on insights from survey research
- We now present three evaluations
 - Quality $\Rightarrow$ RA coders
 - Informative discoveries $\Rightarrow$ Experienced scholars analyzing texts

Evaluating the Performance of Our Method

- Goals:
 - **Validate Claim**: computer-assisted conceptualization outperforms human conceptualization
 - **Demonstrate**: new experimental designs for cluster evaluation
 - **Inject human judgement**: relying on insights from survey research
- We now present three evaluations
 - Quality $\Rightarrow$ RA coders
 - Informative discoveries $\Rightarrow$ Experienced scholars analyzing texts
 - Discovery $\Rightarrow$ You're the judge

Evaluation 1: Cluster Quality

Evaluation 1: Cluster Quality

- What Are Humans Good For?

Evaluation 1: Cluster Quality

- What Are Humans Good For?
 - They can't: keep many documents & clusters in their head

Evaluation 1: Cluster Quality

- What Are Humans Good For?
 - They can't: keep many documents & clusters in their head
 - They can: compare two documents at a time

Evaluation 1: Cluster Quality

- What Are Humans Good For?

- They can't: keep many documents & clusters in their head
- They can: compare two documents at a time
- $\implies$ Cluster quality evaluation: human judgement of document pairs

Evaluation 1: Cluster Quality

- What Are Humans Good For?
 - They can't: keep many documents & clusters in their head
 - They can: compare two documents at a time
 - $\implies$ Cluster quality evaluation: human judgement of document pairs
- Experimental Design to Assess Cluster Quality

Evaluation 1: Cluster Quality

- What Are Humans Good For?
 - They can't: keep many documents & clusters in their head
 - They can: compare two documents at a time
 - $\implies$ Cluster quality evaluation: human judgement of document pairs
- Experimental Design to Assess Cluster Quality
 - automated visualization to choose one clustering

Evaluation 1: Cluster Quality

- What Are Humans Good For?
 - They can't: keep many documents & clusters in their head
 - They can: compare two documents at a time
 - $\Rightarrow$ Cluster quality evaluation: human judgement of document pairs
- Experimental Design to Assess Cluster Quality
 - automated visualization to choose one clustering
 - many pairs of documents

Evaluation 1: Cluster Quality

- What Are Humans Good For?

- They can't: keep many documents & clusters in their head
- They can: compare two documents at a time
- $\Rightarrow$ Cluster quality evaluation: human judgement of document pairs

- Experimental Design to Assess Cluster Quality

- automated visualization to choose one clustering
- many pairs of documents
- for coders: (1) unrelated, (2) loosely related, (3) closely related

Evaluation 1: Cluster Quality

- What Are Humans Good For?

- They can't: keep many documents & clusters in their head
- They can: compare two documents at a time
- $\Rightarrow$ Cluster quality evaluation: human judgement of document pairs

- Experimental Design to Assess Cluster Quality

- automated visualization to choose one clustering
- many pairs of documents
- for coders: (1) unrelated, (2) loosely related, (3) closely related
- $\text{Quality} = \text{mean}(\text{within cluster}) - \text{mean}(\text{between clusters})$

Evaluation 1: Cluster Quality

- What Are Humans Good For?

- They can't: keep many documents & clusters in their head
- They can: compare two documents at a time
- $\Rightarrow$ Cluster quality evaluation: human judgement of document pairs

- Experimental Design to Assess Cluster Quality

- automated visualization to choose one clustering
- many pairs of documents
- for coders: (1) unrelated, (2) loosely related, (3) closely related
- Quality = mean(within cluster) - mean(between clusters)
- Bias results against ourselves by not letting evaluators choose clustering

Evaluation 1: Cluster Quality

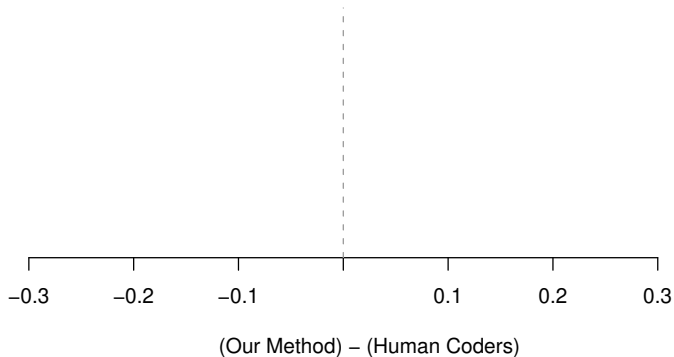
- What Are Humans Good For?

- They can't: keep many documents & clusters in their head
- They can: compare two documents at a time
- $\Rightarrow$ Cluster quality evaluation: human judgement of document pairs

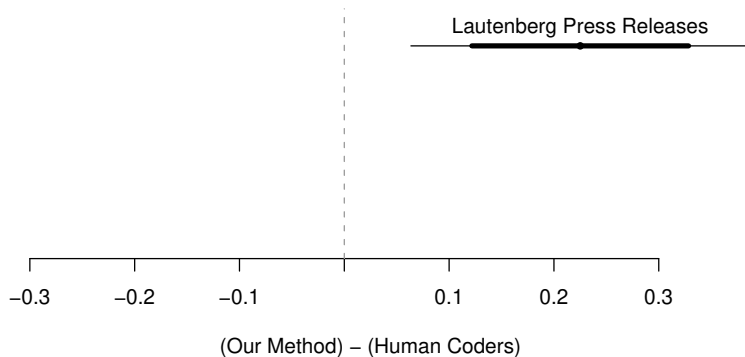
- Experimental Design to Assess Cluster Quality

- automated visualization to choose one clustering
- many pairs of documents
- for coders: (1) unrelated, (2) loosely related, (3) closely related
- Quality = mean(within cluster) - mean(between clusters)
- Bias results against ourselves by not letting evaluators choose clustering

Evaluation 1: Cluster Quality

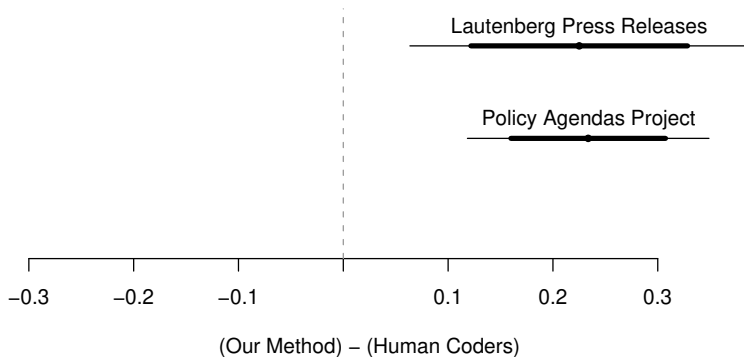


Evaluation 1: Cluster Quality



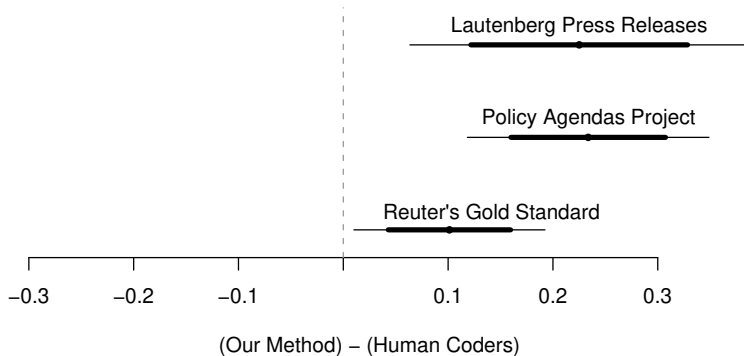
Lautenberg: 200 Senate Press Releases (appropriations, economy, education, tax, veterans, ...)

Evaluation 1: Cluster Quality



Policy Agendas: 213 quasi-sentences from Bush's State of the Union (agriculture, banking & commerce, civil rights/liberties, defense, ...)

Evaluation 1: Cluster Quality



Reuter's: financial news (trade, earnings, copper, gold, coffee, . . .); "gold standard" for supervised learning studies

Evaluation 2: More Informative Discoveries

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)
 - 2 clusterings from each of 2 other methods (varying tuning parameters)

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)
 - 2 clusterings from each of 2 other methods (varying tuning parameters)
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)
 - 2 clusterings from each of 2 other methods (varying tuning parameters)
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)
 - 2 clusterings from each of 2 other methods (varying tuning parameters)
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- User chooses $\Rightarrow$ only care about the one clustering that wins

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)
 - 2 clusterings from each of 2 other methods (varying tuning parameters)
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- User chooses $\Rightarrow$ only care about the one clustering that wins
- Both cases a Condorcet winner:

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)
 - 2 clusterings from each of 2 other methods (varying tuning parameters)
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- User chooses $\Rightarrow$ only care about the one clustering that wins
- Both cases a Condorcet winner:

“Immigration”:

Our Method 1 $\rightarrow$ vMF 1 $\rightarrow$ vMF 2 $\rightarrow$ Our Method 2 $\rightarrow$ K-Means 1 $\rightarrow$ K-Means 2

Evaluation 2: More Informative Discoveries

- Found 2 scholars analyzing lots of textual data for their work
- Created 6 clusterings:
 - 2 clusterings selected with our method (**biased** against us)
 - 2 clusterings from each of 2 other methods (varying tuning parameters)
- Created info packet on each clustering (for each cluster: exemplar document, automated content summary)
- Asked for $\binom{6}{2}=15$ pairwise comparisons
- User chooses $\Rightarrow$ only care about the one clustering that wins
- Both cases a Condorcet winner:

“Immigration”:

Our Method 1 $\rightarrow$ vMF 1 $\rightarrow$ vMF 2 $\rightarrow$ Our Method 2 $\rightarrow$ K-Means 1 $\rightarrow$ K-Means 2

“Genetic testing”:

Our Method 1 $\rightarrow$ {Our Method 2, K-Means 1, K-means 2} $\rightarrow$ Dir Proc. 1 $\rightarrow$ Dir Proc. 2

Evaluation 3: What Do Members of Congress Do?

Evaluation 3: What Do Members of Congress Do?

- David Mayhew's (1974) famous typology

Evaluation 3: What Do Members of Congress Do?

- David Mayhew's (1974) famous typology
 - Advertising

Evaluation 3: What Do Members of Congress Do?

- David Mayhew's (1974) famous typology
 - Advertising
 - Credit Claiming

Evaluation 3: What Do Members of Congress Do?

- David Mayhew's (1974) famous typology
 - Advertising
 - Credit Claiming
 - Position Taking

Evaluation 3: What Do Members of Congress Do?

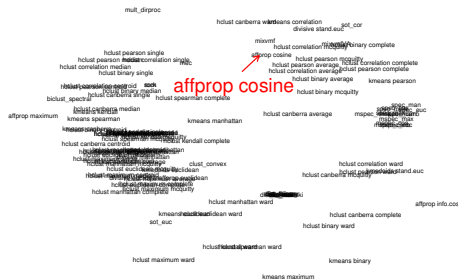
- David Mayhew's (1974) famous typology
 - Advertising
 - Credit Claiming
 - Position Taking
- Data: 200 press releases from Frank Lautenberg's office (D-NJ)

Evaluation 3: What Do Members of Congress Do?

- David Mayhew's (1974) famous typology
 - Advertising
 - Credit Claiming
 - Position Taking
- Data: 200 press releases from Frank Lautenberg's office (D-NJ)
- Apply our method

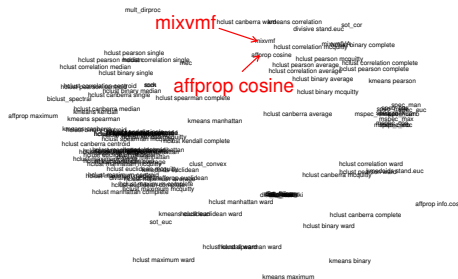


Example Discovery



Red point: a **clustering** by Affinity Propagation-Cosine (Dueck and Frey 2007)

Example Discovery



Red point: a **clustering** by
Affinity Propagation-Cosine
(Dueck and Frey 2007)

Close to:

Mixture of von Mises-Fisher distributions (Banerjee et. al. 2005)



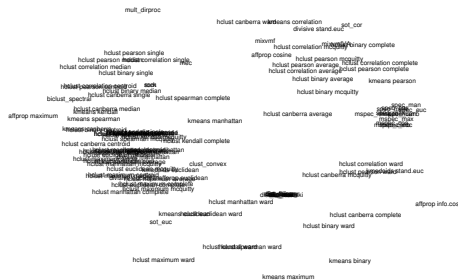


Example Discovery



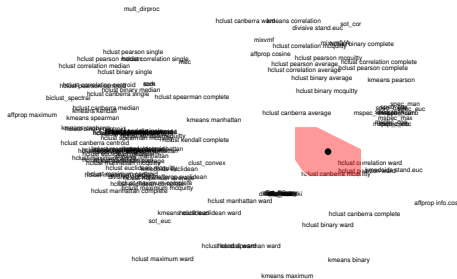
Space between methods:
local cluster ensemble

Example Discovery





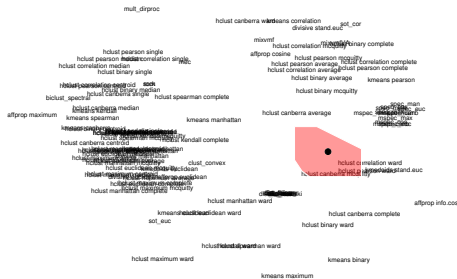
Example Discovery



Mixture:

0.39 Hclust-Canberra-McQuitty

Example Discovery

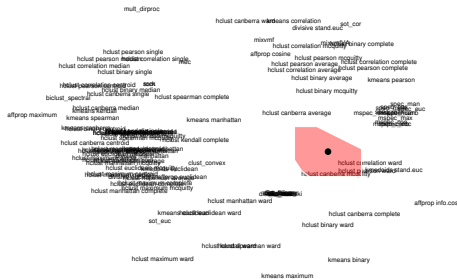


Mixture:

0.39 Hclust-Canberra-McQuitty

0.30 Spectral clustering
Random Walk
(Metrics 1-6)

Example Discovery



Mixture:

0.39 Hclust-Canberra-McQuitty

0.30 Spectral clustering
Random Walk
(Metrics 1-6)

0.13 Hclust-Correlation-Ward

Example Discovery



Mixture:

0.39 Hclust-Canberra-McQuitty

0.30 Spectral clustering
Random Walk
(Metrics 1-6)

0.13 Hclust-Correlation-Ward

0.09 Hclust-Pearson-Ward

Example Discovery



Mixture:

0.39 Hclust-Canberra-McQuitty

0.30 Spectral clustering
Random Walk
(Metrics 1-6)

0.13 Hclust-Correlation-Ward

0.09 Hclust-Pearson-Ward

0.05 Kmediods-Cosine

Example Discovery



Mixture:

0.39 Hclust-Canberra-McQuitty

0.30 Spectral clustering
Random Walk
(Metrics 1-6)

0.13 Hclust-Correlation-Ward

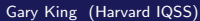
0.09 Hclust-Pearson-Ward

0.05 Kmediods-Cosine

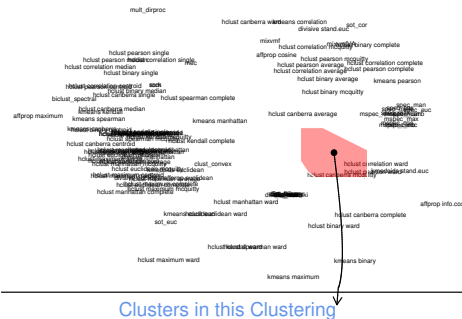
0.04 Spectral clustering

Symmetric
(Metrics 1-6)

Mayhew



Example Discovery

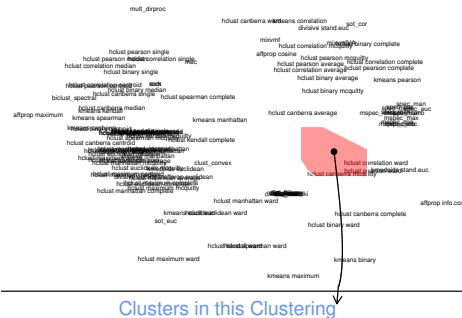


Credit Claiming
Pork

Credit Claiming, Pork:
 “Sens. Frank R. Lautenberg (D-NJ) and Robert Menendez (D-NJ) announced that the U.S. Department of Commerce has awarded a \$100,000 grant to the South Jersey Economic Development District”

Mayhew

Example Discovery



Credit Claiming, Legislation:

“As the Senate begins its recess, Senator Frank Lautenberg today pointed to a string of victories in Congress on his legislative agenda during this work period”

Credit Claiming Pork

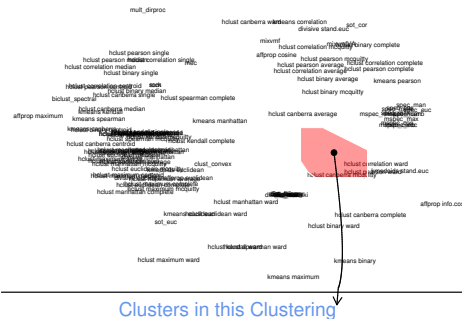
Credit Claiming Legislation

Mayhew

Gary King (Harvard IQSS)

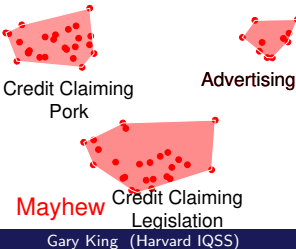
Quantitative Discovery

Example Discovery

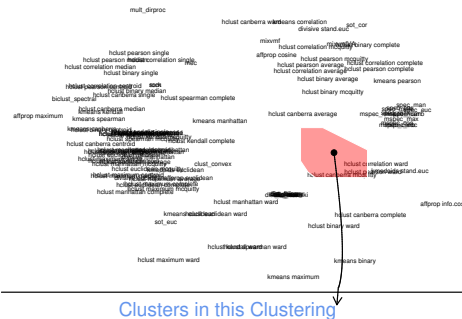


Advertising:

“Senate Adopts
Lautenberg/Menendez Resolution
Honoring Spelling Bee Champion
from New Jersey”

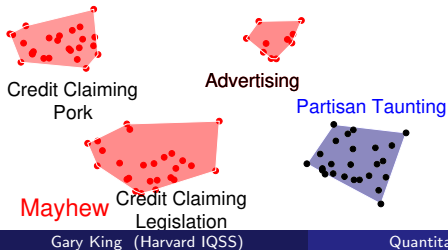


Example Discovery: Partisan Taunting

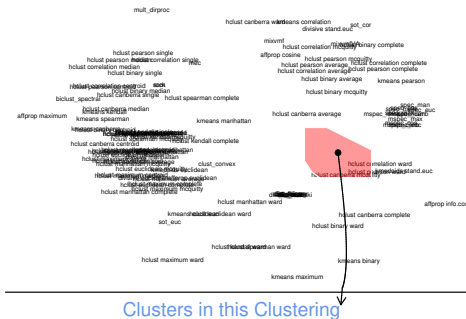


Partisan Taunting:

“Republicans Selling Out Nation on Chemical Plant Security”



Example Discovery: Partisan Taunting



Partisan Taunting:

“Senator Lautenberg’s amendment would change the name of ...the Republican bill...to ‘More Tax Breaks for the Rich and More Debt for Our Grandchildren Deficit Expansion Reconciliation Act of 2006’ ”



Credit Claiming
Pork

Advertising

Partisan Taunting

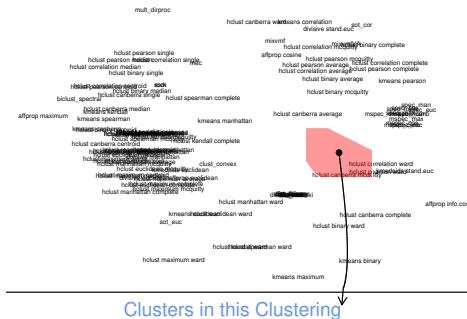
Mayhew

Credit Claiming Legislation

Gary King (Harvard IQSS)

Quantitative Discovery

Example Discovery: Partisan Taunting



Definition: Explicit, public, and negative attacks on another political party or its members

Clusters in this Clustering



Credit Claiming Pork



Advertising



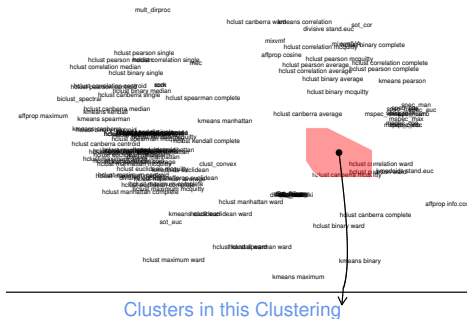
Mayhew

Credit Claiming Legislation



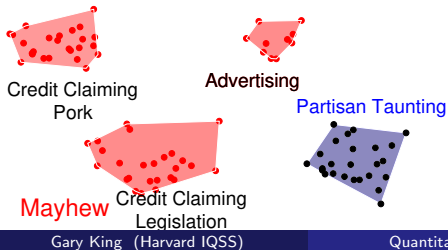
Partisan Taunting

Example Discovery: Partisan Taunting



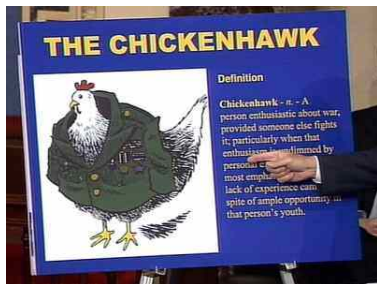
Definition: Explicit, public, and negative attacks on another political party or its members

Taunting ruins deliberation



In Sample Illustration of Partisan Taunting

Taunting ruins deliberation

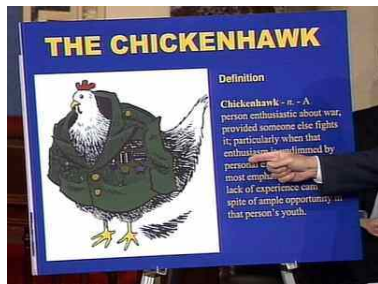


Sen. Lautenberg
on Senate Floor
4/29/04

- “Senator Lautenberg Blasts Republicans as ‘Chicken Hawks’ ”
[Government Oversight]

In Sample Illustration of Partisan Taunting

Taunting ruins deliberation

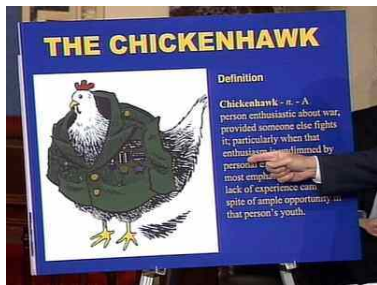


Sen. Lautenberg
on Senate Floor
4/29/04

- “Senator Lautenberg Blasts Republicans as ‘Chicken Hawks’ ” [Government Oversight]
- “The scopes trial took place in 1925. Sadly, President Bush’s veto today shows that we haven’t progressed much since then” [Healthcare]

In Sample Illustration of Partisan Taunting

Taunting ruins deliberation



Sen. Lautenberg
on Senate Floor
4/29/04

- "Senator Lautenberg Blasts Republicans as 'Chicken Hawks' " [Government Oversight]
- "The scopes trial took place in 1925. Sadly, President Bush's veto today shows that we haven't progressed much since then" [Healthcare]
- "Every day the House Republicans dragged this out was a day that made our communities less safe." [Homeland Security]

Out of Sample Confirmation of Partisan Taunting

- Discovered using 200 press releases; 1 senator.

Out of Sample Confirmation of Partisan Taunting

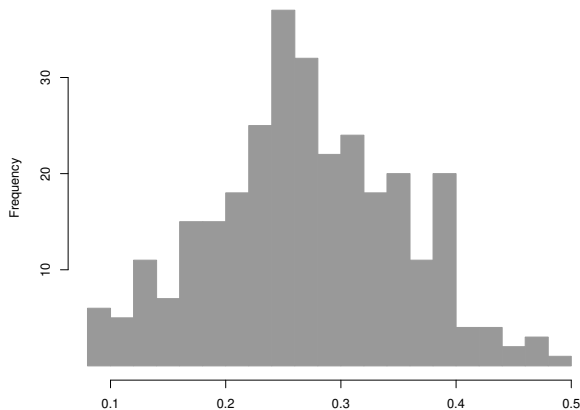
- Discovered using 200 press releases; 1 senator.
- Confirmed using 64,033 press releases; 301 senator-years.

Out of Sample Confirmation of Partisan Taunting

- Discovered using 200 press releases; 1 senator.
- Confirmed using 64,033 press releases; 301 senator-years.
- Apply supervised learning method: measure **proportion of press releases** a senator taunts other party

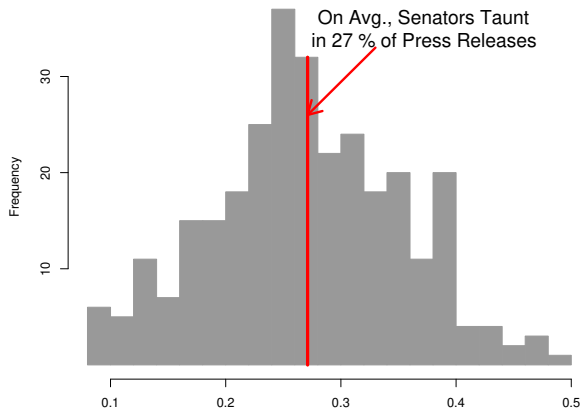
Out of Sample Confirmation of Partisan Taunting

- Discovered using 200 press releases; 1 senator.
- Confirmed using 64,033 press releases; 301 senator-years.
- Apply supervised learning method: measure **proportion of press releases** a senator taunts other party

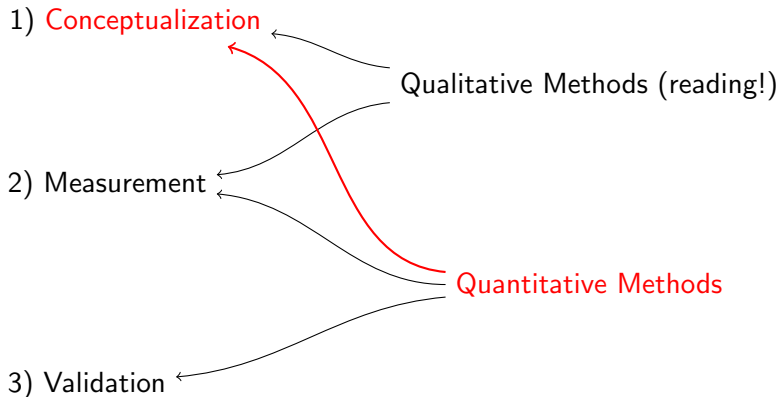


Out of Sample Confirmation of Partisan Taunting

- Discovered using 200 press releases; 1 senator.
- Confirmed using 64,033 press releases; 301 senator-years.
- Apply supervised learning method: measure **proportion of press releases** a senator taunts other party

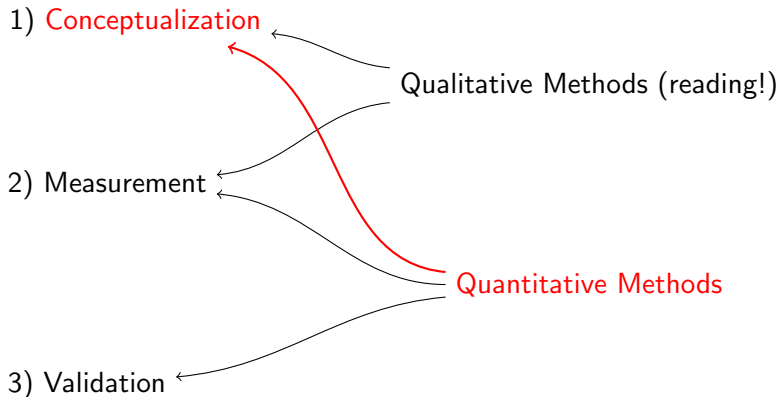


Advancing the Objective of Discovery



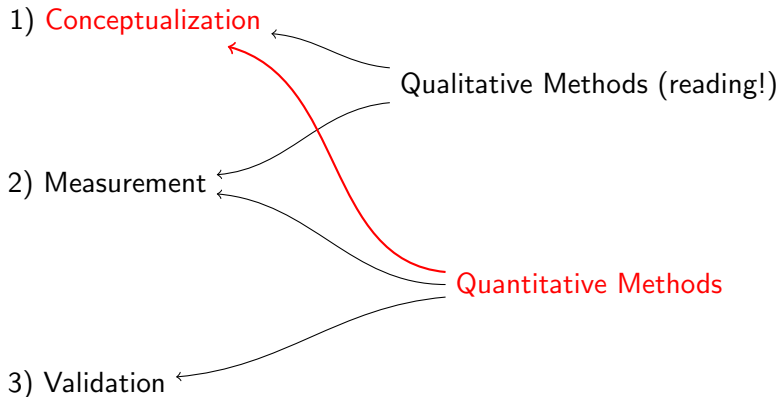
Quantitative methods for conceptualization: aiding **discovery**

Advancing the Objective of Discovery



- Quantitative methods for conceptualization: aiding **discovery**
- Few formal methods designed explicitly for conceptualization

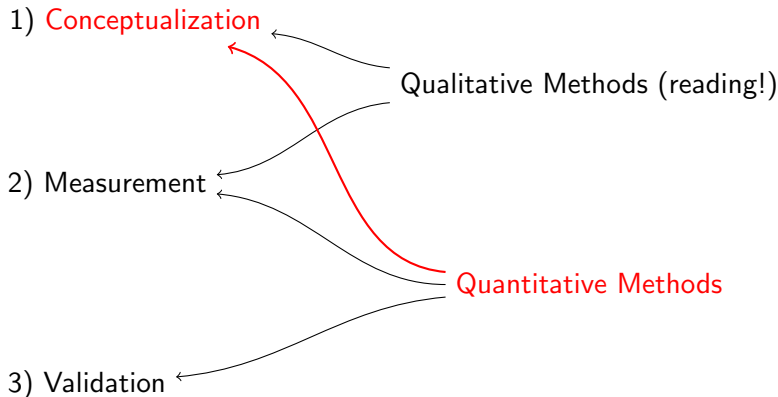
Advancing the Objective of Discovery



Quantitative methods for conceptualization: aiding **discovery**

- Few formal methods designed explicitly for conceptualization
- **Belittled**: “Tom Swift and His Electric Factor Analysis Machine” (Armstrong 1967)

Advancing the Objective of Discovery



Quantitative methods for conceptualization: aiding **discovery**

- Few formal methods designed explicitly for conceptualization
- **Belittled**: “Tom Swift and His Electric Factor Analysis Machine” (Armstrong 1967)
- Evaluation methods measure progress in discovery

For more information (on adding zooming out to the human ability to zoom in)

<http://GKing.Harvard.edu>